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Surgical stapling device with lockout system for preventing actuation in the absence of an installed staple cartridge

Abstract

Surgical stapling instruments having surgical stapling cartridges are disclosed. At least one embodiment includes an end effector that has an elongate channel that is configured to operably support a surgical staple cartridge therein and an anvil that is movably supported on the elongate channel between an open position and closed positions in response to an application of opening and closing motions applied thereto. An anvil lock member cooperates with the anvil to retain the anvil in an open position when a staple cartridge has not been installed in the elongate channel and prevent the anvil from moving to a closed position until the staple cartridge has been seated within the elongate channel. Surgical staple cartridges are configured to move the anvil from an open position to an actuatable position when the cartridge has been installed in the end effector.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5413267	12/1994	Solyntijes et al.	N/A	N/A
5462215	12/1994	Viola et al.	N/A	N/A
5584425	12/1995	Savage et al.	N/A	N/A
5636780	12/1996	Green et al.	N/A	N/A
5704534	12/1997	Huitema et al.	N/A	N/A
5752644	12/1997	Bolanos et al.	N/A	N/A
5762255	12/1997	Chrisman et al.	N/A	N/A
5779130	12/1997	Alesi et al.	N/A	N/A
6964363	12/2004	Wales et al.	N/A	N/A
7000818	12/2005	Shelton, IV et al.	N/A	N/A
7044352	12/2005	Shelton, IV et al.	N/A	N/A
7044353	12/2005	Mastri et al.	N/A	N/A
7111769	12/2005	Wales et al.	N/A	N/A
7143923	12/2005	Shelton, IV et al.	N/A	N/A
7669746	12/2009	Shelton, IV	N/A	N/A
8672209	12/2013	Crainich	N/A	N/A
9078653	12/2014	Leimbach et al.	N/A	N/A

10166025	12/2018	Leimbach et al.	N/A	N/A
10639034	12/2019	Harris	N/A	A61B 17/07207
10736629	12/2019	Shelton, IV et al.	N/A	N/A
11083458	12/2020	Harris et al.	N/A	N/A
12082814	12/2023	Leimbach et al.	N/A	N/A
2004/0232200	12/2003	Shelton, IV	227/176.1	A61B 17/07207
2007/0010838	12/2006	Shelton, IV	606/157	A61B 17/07207
2007/0084896	12/2006	Doll et al.	N/A	N/A
2008/0023522	12/2007	Olson et al.	N/A	N/A
2009/0261143	12/2008	Wixey et al.	N/A	N/A
2011/0204120	12/2010	Crainich	N/A	N/A
2016/0174969	12/2015	Kerr et al.	N/A	N/A
2021/0007743	12/2020	Cropper et al.	N/A	N/A
2021/0282770	12/2020	Fernandes	N/A	A61B 17/07207
2022/0133299	12/2021	Baxter, III	N/A	N/A
2022/0133300	12/2021	Leimbach et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	**Application Date**	**Country**	**CPC**
H05-130998	12/1992	JP	N/A
H05-200038	12/1992	JP	N/A
H09-182757	12/1995	JP	N/A
H06-503244	12/1996	JP	N/A
2161450	12/2000	RU	N/A
61122	12/2006	RU	N/A
WO 2013/148253	12/2012	WO	N/A

OTHER PUBLICATIONS

Extended European Search Report dated Mar. 23, 2020, for Application No. 19219422.3, 8 pages. cited by applicant
Extended European Search Report dated Jul. 23, 2021, for Application No. 21171550.3, 12 pages. cited by applicant
Indian Office Action dated Jun. 16, 2020, for Application No. 7608/DELNP/2014, 5 pages. cited by applicant
International Search Report and Written Opinion dated Jan. 17, 2014, for International Application No. PCT/US2013/031423, 14 pages. cited by applicant
Japanese Notification of Reasons for Refusal dated Mar. 31, 2017, for Application No. 2015-503303, 4 pages. cited by applicant
Russian Office Action dated Mar. 14, 2017, for Application No. 2014143016, 6 pages. cited by applicant
European Examination Report dated Feb. 7, 2019 for Application No. EP 13724428.1, 7 pgs. cited

by applicant
Japanese Search Report by Registered Search Organization dated Jan. 18, 2017 for Application No. JP 2015-503303, 18 pgs. cited by applicant

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application claiming priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 16/230,343, entitled SURGICAL STAPLING DEVICE WITH LOCKOUT SYSTEM FOR PREVENTING ACTUATION IN THE ABSENCE OF AN INSTALLED STAPLE CARTRIDGE, filed Dec. 21, 2018, published as U.S. Patent Application Publication No. 2019/0192161 on Jun. 27, 2019, issued as U.S. Pat. No. 12,082,814 on Sep. 10, 2024, which is a continuation application claiming priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 14/746,305, entitled SURGICAL STAPLING DEVICE WITH LOCKOUT SYSTEM FOR PREVENTING ACTUATION IN THE ABSENCE OF AN INSTALLED STAPLE CARTRIDGE, filed Jun. 22, 2015, which issued Jan. 1, 2019 as U.S. Pat. No. 10,166,025, which is a continuation application claiming priority under 35 U.S.C. § 120 to U.S. patent application Ser. No. 13/429,647, entitled SURGICAL STAPLING DEVICE WITH LOCKOUT SYSTEM FOR PREVENTING ACTUATION IN THE ABSENCE OF AN INSTALLED STAPLE CARTRIDGE, filed Mar. 26, 2012, which issued on Jul. 14, 2015 as U.S. Pat. No. 9,078,653, the entire disclosures of which are hereby incorporated by reference herein.

BACKGROUND
(1) The present invention relates to surgical instruments and, in various embodiments, to surgical cutting and stapling instruments and staple cartridges therefor that are designed to cut and staple tissue.
SUMMARY
(2) In accordance with at least one general form, there is provided a surgical stapling instrument that includes an end effector that has an elongate channel that is configured to operably support a surgical staple cartridge therein. An anvil is movably supported on the elongate channel between an open position and closed positions in response to an application of opening and closing motions applied thereto. An anvil lock member cooperates with the anvil to retain the anvil in an open position when a staple cartridge has not been installed in the elongate channel and prevents the anvil from moving to a closed position until the staple cartridge has been seated within the elongate channel.
(3) In accordance with at least one other general form, there is provided a surgical stapling instrument that includes a handle and an elongate shaft assembly that is operably coupled to the handle. A closure system is operably supported by the handle for generating closing and opening motions in response to actuation of a closure trigger operably supported by the handle. The instrument further includes an elongate channel that is coupled to the elongate shaft assembly. The instrument also includes an anvil that has an anvil mounting portion that is movably supported on the elongate channel between an open position and closed positions in response to applications of opening and closing motions transmitted thereto through the elongate shaft assembly. An anvil lock member is operably supported by the elongate shaft assembly for movable engagement with the anvil mounting portion. A surgical staple cartridge is configured to be seated within the elongate

channel and cooperates with the anvil mounting portion such that when the surgical staple cartridge has not been seated within the elongate channel, the anvil lock member cooperates with the anvil mounting portion to retain the anvil in the open position. In addition, when the surgical staple cartridge is seated within the elongate channel, the staple cartridge moves the anvil mounting portion to a position wherein the anvil may be closed upon application of the closure motions thereto.

(4) In accordance with still another general form, there is provided a surgical staple cartridge for use with a surgical stapling instrument including an end effector with an anvil that is supported in an open position until moved to an actuatable position wherein the anvil is movable to closed positions in response to a closing motion applied thereto by a closure system. In at least one form, the surgical staple cartridge comprises a cartridge body that is sized to be removably seated within a portion of the end effector such that a portion of the cartridge body contacts the anvil to move the anvil from the open position to the actuatable position when seated within the end effector.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The features and advantages of this invention, and the manner of attaining them, will become more apparent and the invention itself will be better understood by reference to the following description of embodiments of the invention taken in conjunction with the accompanying drawings, wherein:

(2) FIG. **1** is a perspective view of a surgical stapling instrument embodiment;

(3) FIG. **2** is an exploded assembly view of the surgical stapling instrument of FIG. **1**;

(4) FIG. **3** is an exploded assembly view of a portion of an articulation assembly embodiment;

(5) FIG. **4** is a partial exploded perspective view of a portion of the handle;

(6) FIG. **5** is a side view of the handle with a handle case removed;

(7) FIG. **6** is a partial exploded perspective view of an end effector and anvil lock embodiment;

(8) FIG. **6**A is a partial exploded perspective view of another end effector and anvil lock member embodiment;

(9) FIG. **7** is a perspective view of an anvil lock member embodiment;

(10) FIG. **7**A is a perspective view of an anvil lock member embodiment of FIG. **6**A;

(11) FIG. **8** is a side elevational view of an end effector embodiment in an open position;

(12) FIG. **9** is a top view of the end effector of FIG. **8**;

(13) FIG. **10** is a bottom view of the end effector depicted in FIGS. **8** and **9**;

(14) FIG. **11** is a partial bottom perspective view of an anvil embodiment;

(15) FIG. **12** is a perspective view of a pivot mount embodiment;

(16) FIG. **13** is a bottom perspective view of the pivot mount embodiment of FIG. **12**;

(17) FIG. **14** is a perspective view of a proximal end portion of a surgical staple cartridge embodiment;

(18) FIG. **15** is a side elevational view of the surgical staple cartridge embodiment depicted in FIG. **14**;

(19) FIG. **16** is a side view of an end effector embodiment prior to seating a staple cartridge in the elongate channel;

(20) FIG. **17** is a cross-sectional view of the end effector depicted in FIG. **16**;

(21) FIG. **18** is a side view of an end effector embodiment of FIGS. **16** and **17** with the anvil in the open position and wherein a surgical staple cartridge is being inserted into the elongate channel;

(22) FIG. **19** is a cross-sectional view of the end effector of FIG. **18**;

(23) FIG. **20** is a side view of the end effector of FIGS. **16**-**19** with the staple cartridge embodiment seated within the elongate channel;

(24) FIG. **21** is a cross-sectional view of the end effector of FIG. **20**;
(25) FIG. **22** is a side elevational view of the end effector of FIGS. **16-22** clamping tissue;
(26) FIG. **23** is a cross-sectional view of the end effector of FIG. **22**;
(27) FIG. **24** is a side elevational view of the end effector of FIGS. **16-23** in a fully clamped position ready to fire;
(28) FIG. **25** is a cross-sectional view of the end effector of FIG. **24**;
(29) FIG. **26** is an exploded assembly view of another surgical stapling instrument embodiment;
(30) FIG. **27** is a perspective view of another pivot mount embodiment;
(31) FIG. **28** is a bottom perspective view of the pivot mount embodiment of FIG. **27**;
(32) FIG. **29** is a partial exploded perspective view of an end effector and another anvil lock member embodiment;
(33) FIG. **30** is a perspective view of another anvil lock member embodiment;
(34) FIG. **31** is a partial side elevational view of a proximal end portion of another surgical staple cartridge embodiment;
(35) FIG. **32** is a perspective view of a proximal end portion of the surgical staple cartridge embodiment of FIG. **31**;
(36) FIG. **33** is a side view of another end effector embodiment prior to seating a staple cartridge in the elongate channel;
(37) FIG. **34** is a cross-sectional view of the end effector depicted in FIG. **33**;
(38) FIG. **35** is a side view of an end effector embodiment of FIGS. **33** and **34** with the anvil in the open position and wherein a surgical staple cartridge is being inserted into the elongate channel;
(39) FIG. **36** is a cross-sectional view of the end effector of FIG. **35**;
(40) FIG. **37** is a side view of the end effector of FIGS. **33-36** with the staple cartridge embodiment seated within the elongate channel;
(41) FIG. **38** is a cross-sectional view of the end effector of FIG. **37**;
(42) FIG. **39** is a side elevational view of the end effector of FIGS. **33-38** clamping tissue;
(43) FIG. **40** is a cross-sectional view of the end effector of FIG. **39**;
(44) FIG. **41** is a side elevational view of the end effector of FIGS. **33-40** in a fully clamped position ready to fire; and
(45) FIG. **42** is a cross-sectional view of the end effector of FIG. **41**.

DETAILED DESCRIPTION

(46) Certain exemplary embodiments will now be described to provide an overall understanding of the principles of the structure, function, manufacture, and use of the devices and methods disclosed herein. One or more examples of these embodiments are illustrated in the accompanying drawings. Those of ordinary skill in the art will understand that the devices and methods specifically described herein and illustrated in the accompanying drawings are non-limiting exemplary embodiments and that the scope of the various embodiments of the present invention is defined solely by the claims. The features illustrated or described in connection with one exemplary embodiment may be combined with the features of other embodiments. Such modifications and variations are intended to be included within the scope of the present invention.
(47) Reference throughout the specification to “various embodiments,” “some embodiments,” “one embodiment,” or “an embodiment”, or the like, means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. Thus, appearances of the phrases “in various embodiments,” “in some embodiments,” “in one embodiment”, or “in an embodiment”, or the like, in places throughout the specification are not necessarily all referring to the same embodiment. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner in one or more embodiments. Thus, the particular features, structures, or characteristics illustrated or described in connection with one embodiment may be combined, in whole or in part, with the features structures, or characteristics of one or more other embodiments without limitation. Such modifications and variations are intended

to be included within the scope of the present invention.

(48) The terms “proximal” and “distal” are used herein with reference to a clinician manipulating the handle portion of the surgical instrument. The term “proximal” referring to the portion closest to the clinician and the term “distal” referring to the portion located away from the clinician. It will be further appreciated that, for convenience and clarity, spatial terms such as “vertical”, “horizontal”, “up”, and “down” may be used herein with respect to the drawings. However, surgical instruments are used in many orientations and positions, and these terms are not intended to be limiting and/or absolute.

(49) Various exemplary devices and methods are provided for performing laparoscopic and minimally invasive surgical procedures. However, the person of ordinary skill in the art will readily appreciate that the various methods and devices disclosed herein can be used in numerous surgical procedures and applications including, for example, in connection with open surgical procedures. As the present Detailed Description proceeds, those of ordinary skill in the art will further appreciate that the various instruments disclosed herein can be inserted into a body in any way, such as through a natural orifice, through an incision or puncture hole formed in tissue, etc. The working portions or end effector portions of the instruments can be inserted directly into a patient's body or can be inserted through an access device that has a working channel through which the end effector and elongate shaft of a surgical instrument can be advanced.

(50) Turning to the Drawings wherein like numerals denote like components throughout the several views, FIGS. **1** and **2** depict a surgical stapling device **10** that is capable of practicing the unique benefits of various embodiments disclosed herein. An exemplary surgical device that has features with which embodiments of the present invention may be effectively employed is disclosed in U.S. Pat. No. 5,704,534, entitled ARTICULATION ASSEMBLY FOR SURGICAL INSTRUMENTS, issued Jun. 6, 1998, the entire disclosure of which is herein incorporated by reference. Various other exemplary surgical stapling device embodiments are described in greater detail in the following U.S. Patents which are each herein incorporated by reference in their respective entireties: U.S. Pat. No. 6,964,363, entitled SURGICAL STAPLING INSTRUMENT HAVING ARTICULATION JOINT SUPPORT PLATES FOR SUPPORTING A FIRING BAR, issued Nov. 15, 2005; U.S. Pat. No. 7,000,818, entitled SURGICAL STAPLING INSTRUMENT HAVING SEPARATE DISTINCT CLOSING AND FIRING MOTIONS, issued Feb. 21, 2006; U.S. Pat. No. 7,044,352, entitled SURGICAL STAPLING INSTRUMENT HAVING A SINGLE LOCKOUT MECHANISM FOR PREVENTION OF FIRING, issued May 16, 2006; U.S. Pat. No. 7,111,769, entitled SURGICAL INSTRUMENT INCORPORATING AN ARTICULATION MECHANISM HAVING ROTATION ABOUT THE LONGITUDINAL AXIS, issued Sep. 26, 2006; and U.S. Pat. No. 7,143,923, entitled SURGICAL STAPLING INSTRUMENT HAVING A FIRING LOCKOUT FOR AN UNCLOSED ANVIL, issued Dec. 5, 2006.

(51) Referring again to FIGS. **1** and **2**, the depicted surgical stapling device **10** includes a handle **20** that is operably connected to an implement portion **22**, the latter further comprising an elongate shaft assembly **30** that is operably coupled to an end effector **200**. The handle **20** includes a pistol grip **24** toward which a closure trigger **152** is pivotally drawn by the clinician to cause clamping, or closing of an anvil **220** toward an elongate channel **210** of the end effector **200**. A firing trigger **102** is farther outboard of the closure trigger **152** and is pivotally drawn by the clinician to cause the stapling and severing of clamped tissue in the end effector **200**.

(52) For example, closure trigger **152** is actuated first. Once the clinician is satisfied with the positioning of the end effector **200**, the clinician may draw back the closure trigger **152** to its fully closed, locked position proximate to the pistol grip **24**. Then, the firing trigger **102** is actuated. The firing trigger **102** springedly returns when the clinician removes pressure. A release button **120** when depressed on the proximal end of the handle **20** releases the locked closure trigger **152**.

(53) Articulation System

(54) The depicted embodiment include an articulation assembly **62** that is configured to facilitate

articulation of the end effector **200** about the elongate axis A-A of the device **10**. Various embodiments, however, may also be effectively employed in connection with non-articulatable surgical stapling devices. As can be seen in FIG. **2**, for example, the elongate shaft assembly **30** includes a proximal closure tube segment **151** that is operably supported by a nozzle **60** that is supported on the handle **20**. The handle **20** may be formed from two handle cases **21**, **23** that operably contain firing and closure systems **100**, **150**. A proximal end portion **153** of the proximal closure tube segment **151** is rotatably supported by the handle **20** to facilitate its selective rotation about the elongate axis A-A. See FIG. **1**. As can also be seen in FIGS. **1** and **2**, in at least one embodiment, a distal end portion **157** of the proximal closure tube segment **151** is coupled to a flexible neck assembly **70**. The flexible neck assembly **70** has first and second flexible neck portions, **72** and **74**, which receive first and second elongate flexible transmission band assemblies **83**, **85**. The first and second transmission band assemblies **83**, **85** have exterior reinforcement band portions **86**, **87**, respectively, extending distally from the structural portions of the bands. Each exterior reinforcement band portion **86**, **87** has a plurality of attachment lugs **88** for securing first and second interior articulation bands **89**, **90**. See FIG. **2**. The transmission band assemblies **83**, **85** may be, for example, composed of a plastic, especially a glass fiber-reinforced amorphous polyamide, sold commercially under the trade name Grivory GV-6H by EMS-American Grilon. In contrast, it may be desired that the interior articulation bands **89**, **90** of the transmission band assemblies **83**, **85** be composed of a metal, advantageously full hard 301 stainless steel or its equivalent. The attachment lugs **88** on the exterior reinforcement band portions **86**, **87** of the transmission bands **83**, **85** are received into and secured within a plurality of lug holes **91** on the corresponding interior articulation band **89**, **90**. At the distal end of the first and second interior articulation band assemblies **89**, **90** there are first and second connectors **92**, **93**. The articulation assembly further comprises distal articulation bands **96** and **97** that are configured to hookingly engage the first and second connectors **92**, **93**, respectively. The articulation bands **96** and **97** have receptacles **98**, **99** to couple the bands **96**, **97** to the end effector **200** as will be discussed in further detail below.

(55) In at least one form, the flexible neck assembly **70** is preferably composed of a rigid thermoplastic polyurethane sold commercially as ISOPLAST grade **2510** by the Dow Chemical Company. As can be seen in FIG. **3**, the flexible neck assembly **70** has first and second flexible neck portions **72**, **74**. These neck portions **72**, **74** are separated by a central longitudinal rib **73**. See FIG. **6**. The neck portions **72**, **74** each have a plurality of neck ribs **75** configured essentially as semi-circular disks. The flexible neck portions **72**, **74** together generally form a cylindrical configuration. A side slot **76** extends through each of the neck ribs **75** to provide a passage through the first and second flexible neck portions **72**, **74** for receiving the interior articulation bands **89**, **90** and exterior reinforcement band portions **86**, **87** of the flexible band assemblies **83**, **85**. In a similar fashion, the central longitudinal rib **73** separating the first and second flexible neck portions **72**, **74** has a central longitudinal slot for providing a passage to receive the stapler actuating members. Extending proximally from the first and second flexible neck portions **72**, **74** are first and second support guide surfaces **77**, **78** for supporting the reciprocating movement of the interior articulation bands **89**, **90** and the exterior reinforcement portions **86**, **87** of the flexible transmission band assemblies **83**, **85**. Extending from the distal end of the flexible neck portions **72**, **74** is a channel guide **79** for guiding the movement of the stapler actuating members into a staple cartridge **300** of the end effector **200** as will be further discussed below.

(56) In at least one form, when the first and second transmission band assemblies **83**, **85** are brought into contact with each other during assembly of the instrument **10**, they form an elongate cylinder which has a longitudinal cavity through it that is concentrically positioned between the band assemblies **83**, **85** for the passage of a firing rod **110**. The proximal ends of the first and second bands have first and second gear racks **94**, **95** which, as will be discussed below, meshingly engage an articulation assembly **62**.

(57) Upon rotation of the articulation assembly **62**, one of the first and second flexible transmission band assemblies is moved forwardly and the other band assembly is moved rearwardly. In response to the reciprocating movement of the band assemblies **83**, **85** within the first and second flexible neck portions **72**, **74** of the flexible neck assembly **70**, the flexible neck assembly **70** bends to provide articulation. As can be seen in FIG. **5**, an articulation assembly **62** includes an actuator **63**, an articulation body **64** and the nozzle **60**. Rotational movement of the actuator **63** causes corresponding rotation of the articulation body **64** within the nozzle **60**. The first and second elongate transmission band assemblies **83**, **85**, consequently reciprocate axially in opposite directions parallel to the longitudinal axis A-A of the endoscopic shaft **30** of the stapling device **10** to cause the remote articulation of the end effector **200** through the flexible neck assembly **70**. The articulation body **64** further includes a drive gear **65** thereon. As can be seen in FIG. **4**, the drive gear **65** has a flared opening **66** through it, and a lower pivot **67**. Within the flared opening **66** of the drive gear **65**, there is a firing rod orifice **68** for receiving the firing rod **110** enabling the firing of staples into the clamped tissue in response to pivotal rotation of the firing trigger **102**. The drive gear **65** is supported for meshing engagement with the first and second drive racks **94**, **95** on the flexible elongate transmission band assemblies **83**, **85** to effect the desired reciprocating movement of the band assemblies **83**, **85**.
(58) As can be seen in FIG. **5**, the nozzle **60** of the articulation assembly **62** has a nozzle body **61**. The nozzle body **61** has an axial bore **69** extending through it for receiving the drive gear **65** of the articulation body **64**. The bore **69** provides a continuous opening axially from the frame into the elongate endoscopic shaft **30** and therefore the firing rod **110** and other operative components of the stapling device **10** can communicate with the end effector **200**. Further details relating to the articulation assembly **62** may be found in U.S. Pat. No. 5,704,534, which has been previously herein incorporated by reference.
(59) Closure System
(60) As will be discussed in further detail below, the end effector **200** comprises an elongate channel **210** that is configured to operably receive a surgical staple cartridge **300**. An anvil **220** is movably supported relative to the elongate channel **210** and is moved from an open position (FIGS. **16** and **17**) to closed positions wherein tissue may be cut and stapled (FIGS. **24** and **25**). The movement of the anvil **220** between open and closed positions is at least partially controlled by a closure system, generally designated as **150**, which, as indicated above, is controlled by the closure trigger **152**. The closure system **150** includes the proximal closure tube segment **151** that operably houses the articulation band assemblies **83**, **85** in the manner discussed above and which is non-movably coupled to the flexible neck assembly **70**.
(61) In various forms, the proximal closure tube segment **151** includes a proximal end portion **153** that axially extends through the bore **69** in the nozzle **60**. The proximal closure tube segment **151** has elongate axial slots **155** therethrough to permit the articulation body **64** to extend therethrough. See FIG. **2**. The slots **155** enable the articulation body **64** to rotate about articulation axis B-B relative to the proximal closure tube segment **151** while facilitating the axial movement of the proximal closure tube segment **151** along axis A-A relative to articulation body **64**. The transmission bands **83**, **85** function as a frame upon which the proximal closure tube segment **151** can axially move. The proximal end **153** of the proximal closure tube segment **151** is rotatably coupled to a closure yoke **154** that is supported within the handle **20** for reciprocating motion therein. See FIGS. **4** and **5**.
(62) The closure trigger **152** has a handle section **156**, a gear segment section **158** and an intermediate section **160**. See FIG. **5**. A bore extends through the intermediate section **160**. A cylindrical support member **162** extending from the second handle housing **23** passes through the bore for pivotably mounting the closure trigger **152** on the handle portion **20**. A proximal end **98** of the closure yoke **154** has a gear rack **164** that is engaged by the gear segment section **158** of the closure trigger **152**. When the closure trigger **152** is moved toward the pistol grip **24** of the handle

portion **20**, the closure yoke **154** and, hence, the proximal closure tube segment **151** move distally, compressing a spring **166** that biases the closure yoke **152** proximally.

(63) In at least one form, the closure system **150** further includes a distal closure tube segment **170** that is non-movably coupled to the channel guide portion **79** of the flex neck assembly **70** by attachment tabs **72**, **74**. See FIGS. **9** and **10**. The distal closure tube segment **170** has an opening **176** therein that is adapted to interface with an upstanding tab **224** formed on the anvil **220** as will be discussed in further detail below. Thus, axial movement of the proximal closure tube segment **151** results in axial movement of the flex neck assembly **70**, as well as the distal closure tube segment **170**. For example, distal movement of the proximal closure tube segment **151** effects pivotal translation movement of the anvil **220** distally and toward the elongate channel **210** of the end effector **200** and proximal movement effects opening of the anvil **220** as will be discussed in further detail below.

(64) Firing System

(65) In at least one form, the surgical instrument **10** further includes a firing system, generally designated as **100**, for applying firing motions to the firing rod **110** in response to actuation of the firing trigger **102**. In at least one form, the firing system **100** further includes a drive member **104** that has first and second gear racks **105**, **106** thereon. A first notch **109** is provided on the drive member **105** intermediate the first and second gear racks **105**, **106**. During return movement of the firing trigger **102**, a tooth **112** on the firing trigger **102** engages with the first notch **109** for returning the drive member **104** to its initial position after staple firing. A second notch **114** is located at a proximal end of the firing rod **110** for locking the firing rod **110** to an upper latch arm **122** of the release button **120** in its unfired position. The firing system **150** further includes first and second integral pinion gears **111**, **113**. The first integral pinion gear **111** is engaged with a drive rack **115** provided on the firing rod **110**. The second integral pinion gear **113** is engaged with the first gear rack **105** on the drive member **104**. The first integral pinion gear **111** has a first diameter and the second integral pinion gear **113** has a second diameter which is smaller than the first diameter.

(66) In various embodiments, the firing trigger **102** is provided with a gear segment section **103**. The gear segment section **103** engages the second gear rack **106** on the drive member **104** such that motion of the firing trigger **102** causes the drive member **104** to move back and forth between first and second drive positions. In order to prevent staple firing before tissue clamping has occurred, the upper latch arm **122** on the release button **120** is engaged with the second notch **114** on the drive rack **115** such that the firing rod **110** is locked in its proximal-most position. When the upper latch arm **122** falls into a recess in the closure yoke, the upper latch arm **122** disengages with the second notch **114** to permit distal movement of the firing rod **110**. Because the first gear rack **105** on the drive member **104** and the drive rack **115** on the firing rod **110** are engaged, movement of the firing trigger **102** causes the firing rod **110** to reciprocate between a first reciprocating position and a second reciprocating position. Further details concerning various aspects of the firing system **150** may be gleaned from reference to U.S. Pat. No. 7,000,818 which has been herein incorporated by reference in its entirety.

(67) As can be seen in FIG. **3**, various embodiments, the distal end **117** of the firing rod **110** is rotatably received within a firing bar mounting yoke **118**. The firing bar mounting yoke **118** has a slot **119** for hookingly receiving a hook **132** formed on a proximal end of a knife bar **130**. In addition, as shown in FIG. **3**, a support bar **140** is supported for axial movement between the first and second support guide surfaces **77**, **78** of the flex neck assembly **70**. The support bar **140** has a slot **142** that is configured to permit the knife bar **130** to slidably pass therethrough. The metal knife bar **130** has a tissue cutting edge **134** formed on its distal end and is configured to operably interface with a wedge sled operably supported within a surgical staple cartridge **300**.

(68) End Effector

(69) As discussed above, in at least one form, an end effector **200** includes an elongate channel **210**

that is configured to operably support a surgical staple cartridge **300** therein. As shown in FIGS. **2** and **6**, the elongate channel **210** has a proximal end portion that includes two spaced mounting tabs **212** that are configured to be engaged by the hooks **998**, **99** on the distal ends of the articulation bands **96**, **97**. Thus, the reciprocating motions of the articulation bands **96**, **97** cause the elongate channel **210** to articulate relative to the flex neck assembly **70**. As further indicated above, the end effector **200** also includes an anvil **220**. In at least one form, the anvil **220** is fabricated from, for example, 416 Stainless Steel Hardened and Tempered RC35 Min (or similar material) and has a staple-forming undersurface **222** thereon that is configured for confronting engagement with the staple cartridge **300** when mounted in the elongate channel **210**. The anvil **220** is formed with a proximally extending mounting portion **223** that includes two trunnion walls **226**, **228** that each has a trunnion **30** protruding therefrom. See FIG. **11**. In addition, formed on the underside **232** of the mounting portion **223** is a downwardly protruding pivot tab **234** that has a slot **236** extending therethrough that is configured to receive and support the knife bar **130** as it is axially advanced through the end effector **200** during cutting and stapling. In addition, the anvil opening tab **224** is formed on the mounting portion **223** such that it can operably interface with the opening **176** in the distal closure tube segment **170** as will be further discussed below. As can be seen in FIGS. **16-25**, the anvil trunnions **230** are configured to be movably received in corresponding trunnion slots **214** formed in the proximal end of the elongate channel **210**. Each trunnion slot **214** has an arcuate segment **216** that communicates with a locking notch **218**.

(70) To facilitate pivotal travel of the anvil mounting portion **223** relative to the elongate channel **210**, various embodiments include a pivot mount **240**. As can be in FIGS. **12** and **13**, one form of a pivot mount **240** has a body portion **242** that is configured to be attached to the elongate channel **210**. For example, the body portion **242** may be formed with two opposed attachment tabs **243** that are configured to retainingly engage tab openings **211** (FIG. **6**) formed in the elongate channel **210**. In addition, the pivot mount **240** has a proximally extending foot portion **244** that has a retainer lug **245** protruding therefrom that is configured to be received in a corresponding opening **211** in the elongate channel **210**. See FIG. **17**. The pivot mount **240** may be fabricated from, for example, Vectra A435 Liquid Crystal Polymer-natural or similar materials. As can be further seen in FIGS. **12** and **13**, the body portion **242** has an upstanding central portion **246** that has a slot **247** extending therethrough for axially receiving the knife bar **130**. The central portion **246** provides lateral support to the knife bar **130** as it is driven through tissue clamped within the end effector **200**. Various embodiments of the pivot mount **240** further include rocker surfaces **248** formed on each side of the central portion **246** for pivotally receiving the trunnion walls **226**, **228** of the anvil **220** thereon.

(71) Anvil Lockout System

(72) Various embodiments include a unique and novel anvil lockout system **250** that prevents closure of the anvil **220** when a staple cartridge **300** has not been properly installed in the elongate channel **210**. Referring to FIGS. **6** and **7**, for example, an embodiment of an anvil lockout system **250** includes a movable anvil lock member **260** that is movable in response to contact by a portion or portions of a staple cartridge **300** as will be discussed in further detail below. In at least one form, the anvil lock member **260** comprises a body portion **262** that has a distally protruding central support tab **264** formed thereon. A slot **266** extends through body portion **262** and the central support tab **264** to enable the knife bar **130** to pass therethrough. The body portion **262** further includes proximally extending mounting bar **268** that is configured to be slidably received within a corresponding mounting opening **270** in the channel guide **79** of the flex neck assembly **70**. In addition, a biasing member in the form of, for example, a coil spring **269** is supported within the opening **270** to bias the anvil lock member **260** in the distal direction “DD”. See FIG. **16**. When the anvil **220** is mounted to the elongate channel **210**, the trunnions **230** are received within their corresponding trunnion slots **214** in the elongate channel **210**, the central support tab **264** of the anvil lock member **260** is received between the trunnion walls **226**, **228** to further provide support

to the anvil **220**. The body portion **262** of the anvil lock member **260** is further formed with two cam surfaces **263** configured to engage the proximal end surfaces **227**, **229** of the trunnion walls **226**, **228**. See FIGS. **6** and **7**. Various embodiments of the anvil lock member may be fabricated from, for example, Vectra A435 Liquid Crystal Polymer-natural or similar materials.

(73) FIGS. **6**A and **7**A illustrate an alternative anvil lock member **260′** that is movable in response to contact by a portion or portions of a staple cartridge **300**. In at least one form, the anvil lock member **260′** comprises a body portion **262** that has a distally protruding central support tab **264** formed thereon. A slot **266** extends through body portion **262** and the central support tab **264** to enable the knife bar **130** to pass therethrough. The body portion **262** further includes proximally extending mounting bar **268′** that is configured to be slidably and retainably received within a corresponding mounting opening **270′** in the channel guide **79′** of the flex neck assembly **70′**. In addition, a biasing member in the form of, for example, a coil spring **269** is supported within the opening **270′** to bias the anvil lock member **260′** in the distal direction “DD”. The anvil lock member **260′** otherwise operates in the same manner as anvil lock member **260**. When the anvil **220** is mounted to the elongate channel **210**, the trunnions **230** are received within their corresponding trunnion slots **214** in the elongate channel **210**, the central support tab **264** of the anvil lock member **260′** is received between the trunnion walls **226**, **228** to further provide support to the anvil **220**. The body portion **262** of the anvil lock member **260** is further formed with two cam surfaces **263** configured to engage the proximal end surfaces **227**, **229** of the trunnion walls **226**, **228**. The distal closure tube segment **170′** operates in the same manner as the distal closure tube segment **170** described above.

(74) Surgical Staple Cartridge

(75) Various embodiments include a unique and novel surgical staple cartridge **300** that is configured to interact with the anvil lockout system **250** when installed in the elongate channel **210**. As can be seen in FIGS. **14** and **15**, in at least one form, the surgical staple cartridge **300** includes a cartridge body **302** that may be fabricated from, for example, Vectra A435, 20% PTFE/15% GF-natural. The cartridge body **302** is sized and shaped to be received within the elongate channel **210**. In at least one form, the cartridge body **302** is configured to be seated in the elongate channel **210** such that is removably retained therein. The cartridge body **302** may be formed with a centrally disposed slot **304** therein for receiving the knife bar **130**. On each side of the slot **304**, there is provided rows **306**, **308**, **310** of staple openings **312** that are configured to support a surgical staple therein. In the depicted embodiment, three rows **306**, **308**, **310** are provided on each side of the slot **304**. The surgical staples may be supported on staple drivers that are movably supported within the staple openings **312**. Also supported within the staple cartridge body **302** is a wedge sled that is configured for axial movement through the cartridge body **302** when contacted by the cutting bar. The wedge sled is configured with wedge-shaped driving members that contact the staple drivers and drive the drivers and their corresponding staples toward the closed anvil as the wedge sled is driven distally through the cartridge body **302**. Examples of staple driver arrangements and wedge sled arrangements that may be employed are described in further detail in U.S. Pat. No. 7,669,746, the entire disclosure which is herein incorporated by reference. In various embodiments, to facilitate installation of the wedge sled and drivers in the cartridge body **302**, metal cartridge pans **314**, **316** may be attached to the cartridge body **302** as shown in FIGS. **14** and **15**. The cartridge pans **314** and **316** serve to retain the wedge sled and drivers within the cartridge body **302**.

(76) In various embodiments, the cartridge body **302** additionally has at least one release member formed thereon that protrudes in the proximal direction. In the embodiment depicted in FIG. **14**, two release members **320** are formed on the proximal end **319** of the cartridge **300**. The release members **320** each have a wedge shape that defines a sloped pivot surface **321** that are configured to pivotally support a portion of the anvil mounting portion **223** thereon.

(77) Installation of a Staple Cartridge

(78) An understanding of the operation of a anvil lockout system may be gleaned from reference to FIGS. **16-25**. FIGS. **16** and **17** illustrate the position of the anvil **220** relative to the elongate channel **210** prior to installing a staple cartridge **300**. When in that "unloaded" and open position, the anvil lock member **260** is biased in the distal direction by spring **269** such that the cam surfaces **263** on the anvil lock member **260** are in contact with the end surfaces **227**, **229** of the trunnion walls **226**, **228**. The anvil lock member **260** pushes the anvil mounting portion **223** in the distal direction "DD" such that the trunnions **230** are seated in their respective locking notch **218**. The cam surfaces **263** on the anvil lock member **260**, in cooperation with the end wall surfaces **227**, **229**, also serve to pivot and retain the anvil in the open position as shown in FIGS. **16** and **17**. As can be seen in FIG. **16**, when in that position, the trunnion walls **226**, **228** are supported on the rocker surfaces **248** on the pivot mount **240**. When in that position, the surgeon cannot close the anvil **220** by actuating the closure trigger **152** to advance the distal closure tube **170**. Because the closure tube segments cannot be advanced distally to close the anvil **220**, the closure trigger **152** cannot be actuated to its fully closed position whereby the firing trigger **102** may be actuated. Thus, when no cartridge **300** is present, the end effector **200** may not be actuated.
(79) FIGS. **18** and **19** illustrate the initial insertion of the staple cartridge **300** into the elongate channel **210**. FIGS. **20** and **21** illustrate the end effector **200** after the staple cartridge **300** has been fully seated in the elongate channel **210**. As can be seen in FIG. **20** for example, when the cartridge **300** has been fully seated, the release members **320** on the cartridge **300** engage the trunnion walls **226**, **228** and serve to move the anvil mounting portion **223** in a proximal direction "PD" such that the trunnion walls **226**, **228** now pivotally rest on the release members **320**. As can be seen in FIG. **21**, when in that position, the anvil mounting portion **223** has moved proximally such that the trunnions **230** are moved out of their respective locking notches **218** and into the bottom of the arcuate slot segment **216** into an "actuatable" position whereby the anvil **220** may be pivoted closed by actuating the closure trigger **152**.
(80) When the device **10** is in the starting position and the staple cartridge **300** has been loaded into the elongate channel as described above, both of the triggers **152**, **102** are forward and the anvil **220** has been moved to the actuatable position, such as would be typical after inserting the loaded end effector **200** through a trocar or other opening into a body cavity. The instrument **10** is then manipulated by the clinician such that tissue "T" to be stapled and severed is positioned between the staple cartridge **300** and the anvil **200**, as depicted in FIGS. **22** and **23**. As discussed above, movement of the closure trigger **152** toward the pistol grip **24** causes the proximal closure tube segment **151**, the flex neck assembly **70** and the distal closure tube segment **170** to move distally. As the distal closure tube segment **170** moves distally, it contacts a closure ledge **221** on the anvil **220**. Pressure from the tissue captured between the anvil **220** and the staple cartridge **300** serves to move the anvil **220** such that the trunnions **230** are positioned to move within the arcuate trunnion slot segments **216**. The surgeon may pivot the anvil **220** relative to the staple cartridge **300** to manipulate and capture the desired tissue "T" in the end effector **200**. As the distal closure tube segment **170** contacts the closure ledge **221**, the anvil **220** is pivoted towards a clamped position. The retracted knife bar **130** does not impede the selective opening and closing of the anvil **220**.
(81) Once the desired tissue "T" has been positioned between the anvil **220** and the cartridge **300**, the clinician moves the closure trigger **152** proximally until positioned directly adjacent to the pistol grip **24**, locking the handle **20** into the closed and clamped position. As can be seen in FIG. **25**, when in the fully clamped position, the anvil trunnions **230** are located in the upper end of the arcuate slot portion **216** and the anvil tab **224** is received within the opening **176** in the distal closure tube segment **170**. After tissue clamping has occurred, the clinician moves the firing trigger **102** proximally causing the knife bar **130** to move distally into the end effector **200**. In particular, the knife bar **130** moves through the slot **236** in the pivot tab portion **234** of the anvil **220** and into the slot **304** in the cartridge body **302** to contact the wedge sled operably positioned within the staple cartridge **300**. As the knife bar **130** is driven distally, it cuts the tissue T and drives the wedge

sled distally which causes the staples to be sequentially fired into forming contact with the staple-forming undersurface **222** of the anvil **220**. The clinician continues moving the firing trigger **102** until brought proximal to the closure trigger **152** and pistol grip **24**. Thereby, all of the ends of the staples are bent over as a result of their engagement with the anvil **220**. The cutting edge **132** has traversed completely through the tissue T. The process is complete by releasing the firing trigger **102** and by then depressing the release button **120** while simultaneously squeezing the closure trigger **152**. Such action results in the movement of the distal closure tube segment **170** in the proximal direction “D”. As the anvil tab **224** is engaged by the opening **176** in the distal closure tube segment **170** it causes the anvil to pivot open. The end surfaces **227**, **229** again contact the pusher surfaces **263** on the anvil lock member **260** to pivot the anvil to the open position shown in FIGS. **20** and **21** to enable the spent cartridge **300** to be removed from the elongate channel **210**.
(82) FIGS. **26-42** illustrate an alternative surgical stapling instrument **10′** that is similar in construction and operation to surgical stapling instrument **10** except for the differences discussed below. This embodiment, for example, employs the pivot mount **240′** illustrated in FIGS. **29** and **30**. As can be seen in FIGS. **27** and **28** one form of a pivot mount **240′** has a body portion **242′** that is configured to be attached to the elongate channel **210**. For example, the body portion **242′** may be formed with two opposed attachment tabs **243′** that are configured to retainingly engage tab openings **211** (FIG. **26**) formed in the elongate channel **210**. In addition, the pivot mount **240′** has a proximally extending foot portion **244′** that has a slot **247′** extending therethrough for axially receiving the knife bar **130**. Various embodiments of the pivot mount **240′** further include rocker surfaces **248′** formed on the body portion **242′** for pivotally receiving the trunnion walls **226**, **228** of the anvil **220** thereon.
(83) This embodiment also includes an anvil lockout system **250′** that prevents closure of the anvil **220** when a staple cartridge **300′** has not been properly installed in the elongate channel **210**. Referring to FIGS. **29** and **30**, for example, an embodiment of an anvil lockout system **250′** includes an anvil lock member **400** that is configured to contact the anvil mounting portion **223** as will be discussed in further detail below. In at least one form, the anvil lock member **400** comprises a leaf spring **402** that has a slot **404** therein for accommodating the knife bar **130**. The leaf spring **402** is configured for attachment to the channel guide **79″** of the flex neck assembly **70″**.
(84) As can be seen in FIGS. **31** and **32**, in at least one form, the surgical staple cartridge **300′** includes a cartridge body **302′** that is similar to the surgical staple cartridge **300** described above, except for the differences discussed below. FIG. **29** depicts a wedge sled **360** that is supported within the cartridge body **302′** in the manner described above. In this embodiment, the proximal end portion **303** of the cartridge body **302′** is configured to contact a portion of the anvil mounting portion **223** and urge the anvil **220** proximally when the cartridge body **302′** is seated within the elongate channel **210**.
(85) An understanding of the operation of a anvil lockout system **250′** may be gleaned from reference to FIGS. **33-42**. FIGS. **33** and **34** illustrate the position of the anvil **220** relative to the elongate channel **210** prior to installing a staple cartridge **300′**. When in that “unloaded” position, the anvil lock member **400** has engaged the upper surface of the anvil support portion **223** such that the anvil **220** is pivoted to the open position on the rocker surfaces **248′** on the pivot mount **140′**. When in that position, the trunnions **230** are seated in their respective locking notch **218**. When in that position, the surgeon cannot close the anvil **220** by actuating the closure trigger **152** to advance the distal closure tube **170′**. Because the closure tube segments cannot be advanced distally to close the anvil **220**, the closure trigger **152** cannot be actuated to its fully closed position whereby the firing trigger **102** may be actuated. Thus, when no cartridge **300′** is present, the end effector **200** may not be actuated.
(86) FIGS. **35** and **36** illustrate the initial insertion of the staple cartridge **300′** into the elongate channel **210**. FIGS. **37** and **38** illustrate the end effector **200** after the staple cartridge **300′** has been fully seated in the elongate channel **210**. As can be seen in FIG. **37** for example, when the cartridge

300′ has been fully seated, the proximal end portion **303** on the cartridge **300′** engages the trunnion walls **226**, **228** and serves to move the anvil mounting portion **223** in a proximal direction "PD" such that the trunnions are moved out of their respective locking notch **218** and into an actuatable position the bottom of the arcuate slot segment **216**. The anvil **220** is now in position to be pivoted closed by actuating the closure trigger **152**.

(87) When the device **10′** is in the starting position and the staple cartridge **300′** has been loaded into the elongate channel **210** as described above, both of the triggers **152**, **102** are forward and the anvil **220** is open and in the actuatable position, such as would be typical after inserting the loaded end effector **200** through a trocar or other opening into a body cavity. The instrument **10′** is then manipulated by the clinician such that tissue "T" to be stapled and severed is positioned between the staple cartridge **300′** and the anvil **220**, as depicted in FIGS. **39** and **40**. As discussed above, movement of the closure trigger **152** toward the pistol grip **24** causes the proximal closure tube segment **151**, the flex neck assembly **70″** and the distal closure tube segment **170″** to move distally. As the distal closure tube segment **170′** moves distally, it contacts a closure ledge **221** on the anvil **220**. Pressure from the tissue captured between the anvil **220** and the staple cartridge **300′** serves to move the anvil **220** such that the trunnions **230** are positioned to move within the arcuate trunnion slot segments **216**. The surgeon may pivot the anvil **220** relative to the staple cartridge to manipulate and capture the desired tissue "T" in the end effector **200**. As the distal closure tube segment **170″** contacts the closure ledge **221**, the anvil **220** is pivoted towards a clamped position. The retracted knife bar **130** does not impede the selective opening and closing of the anvil **220**.

(88) Once the desired tissue "T" has been positioned between the anvil **220** and the cartridge **300′**, the clinician moves the closure trigger **152** proximally until positioned directly adjacent to the pistol grip **24**, locking the handle **20** into the closed and clamped position. As can be seen in FIG. **42**, when in the fully clamped position, the anvil trunnions **230** are located in the upper end of the arcuate slot portion **216** and the anvil tab **224** is received within the opening **176** in the distal closure tube segment **170″**. After tissue clamping has occurred, the clinician moves the firing trigger **102** proximally causing the knife bar **130** to move distally into the end effector **200**. In particular, the knife bar **130** moves through the slot **236** in the pivot tab portion **234** of the anvil **220** and into the slot **304** in the cartridge body **302′** to contact the wedge sled **360** operably positioned in therein. As the knife bar **130** is driven distally, it cuts the tissue T and drives the wedge sled **360** distally which causes the staples to be sequentially fired into forming contact with the staple-forming undersurface **222** of the anvil **220**. The clinician continues moving the firing trigger **102** until brought proximal to the closure trigger **152** and pistol grip **24**. Thereby, all of the ends of the staples are bent over as a result of their engagement with the anvil **220**. The cutting edge **132** has traversed completely through the tissue T. The process is complete by releasing the firing trigger **102** and by then depressing the release button **120** while simultaneously squeezing the closure trigger **152**. Such action results in the movement of the distal closure tube segment **170″** in the proximal direction "D". As the anvil tab **224** is engaged by the opening **176** in the distal closure tube segment **170″**, it causes the anvil **220** to pivot open. The anvil lock member **400** applies a biasing force to the upper surface of the trunnion walls of the anvil mounting portion **223** and serves to pivot the anvil to the open position shown in FIGS. **33** and **34** to enable the spent cartridge **300′** to be removed from the elongate channel **210**.

(89) The various unique and novel features of the above-described embodiments serve to prevent the end effector from being closed when a surgical staple cartridge is not present or has not been properly seated within the elongate channel. When the anvil is in the locked position wherein the anvil trunnions are retained in their respective locking notches, the anvil is retained in the open position. When in the open position, the end effector cannot be inadvertently inserted through a trocar. Because a full closure stroke is prevented, the firing system cannot be actuated. Thus, even if the clinician attempts to actuate the firing trigger, the device will not fire. Various embodiments also provide the clinician with feedback indicating that a cartridge is either not present or has not

been properly installed in the elongate channel.
(90) The devices disclosed herein can be designed to be disposed of after a single use, or they can be designed to be used multiple times. In either case, however, the device can be reconditioned for reuse after at least one use. Reconditioning can include any combination of the steps of disassembly of the device, followed by cleaning or replacement of particular pieces, and subsequent reassembly. In particular, the device can be disassembled, and any number of the particular pieces or parts of the device can be selectively replaced or removed in any combination. Upon cleaning and/or replacement of particular parts, the device can be reassembled for subsequent use either at a reconditioning facility, or by a surgical team immediately prior to a surgical procedure. Those skilled in the art will appreciate that reconditioning of a device can utilize a variety of techniques for disassembly, cleaning/replacement, and reassembly. Use of such techniques, and the resulting reconditioned device, are all within the scope of the present application.
(91) Preferably, the invention described herein will be processed before surgery. First, a new or used instrument is obtained and if necessary cleaned. The instrument can then be sterilized. In one sterilization technique, the instrument is placed in a closed and sealed container, such as a plastic or TYVEK bag. The container and instrument are then placed in a field of radiation that can penetrate the container, such as gamma radiation, x-rays, or high-energy electrons. The radiation kills bacteria on the instrument and in the container. The sterilized instrument can then be stored in the sterile container. The sealed container keeps the instrument sterile until it is opened in the medical facility.
(92) Any patent, publication, or other disclosure material, in whole or in part, that is said to be incorporated by reference herein is incorporated herein only to the extent that the incorporated materials does not conflict with existing definitions, statements, or other disclosure material set forth in this disclosure. As such, and to the extent necessary, the disclosure as explicitly set forth herein supersedes any conflicting material incorporated herein by reference. Any material, or portion thereof, that is said to be incorporated by reference herein, but which conflicts with existing definitions, statements, or other disclosure material set forth herein will only be incorporated to the extent that no conflict arises between that incorporated material and the existing disclosure material.
(93) While this invention has been described as having exemplary designs, the present invention may be further modified within the spirit and scope of the disclosure. This application is therefore intended to cover any variations, uses, or adaptations of the invention using its general principles. Further, this application is intended to cover such departures from the present disclosure as come within known or customary practice in the art to which this invention pertains.

Claims

1. A surgical stapling instrument comprising: (a) a shaft extending along a longitudinal axis to a distal end; and (b) an end effector operatively coupled with the shaft, wherein the end effector includes: (i) a first jaw including: (A) an anvil, (B) at least one trunnion, and (C) at least one first abutment surface, and (ii) a second jaw configured to cooperate with the first jaw to clamp tissue, wherein the second jaw is configured to support a stapling assembly, wherein the second jaw includes: (A) at least one locking notch configured to receive the at least one trunnion, and (B) at least one second abutment surface configured to engage the at least one first abutment surface when the at least one trunnion is received within the at least one locking notch to thereby prevent closure of the first jaw toward the second jaw.
2. The surgical stapling instrument of claim 1, wherein the first jaw includes at least one trunnion wall, wherein the at least one trunnion protrudes from the at least one trunnion wall, wherein the at least one trunnion wall includes the at least one first abutment surface.

3. The surgical stapling instrument of claim 1, wherein the second jaw includes a pivot mount, wherein the pivot mount includes the at least one second abutment surface.
4. The surgical stapling instrument of claim 3, wherein the pivot mount includes an upstanding central portion having a slot extending therethrough for axially receiving a knife bar.
5. The surgical stapling instrument of claim 4, wherein the at least one second abutment surface includes a pair of second abutment surfaces positioned on opposite sides of the central portion.
6. The surgical stapling instrument of claim 3, wherein the second jaw includes an elongate channel configured to receive the stapling assembly, wherein the pivot mount is attached to the elongate channel.
7. The surgical stapling instrument of claim 6, wherein the pivot mount includes at least one attachment tab, wherein the elongate channel includes at least one opening configured to receive the at least one attachment tab.
8. The surgical stapling instrument of claim 6, wherein the pivot mount includes a retainer lug, wherein the elongate channel includes at least one opening configured to receive the retainer lug.
9. The surgical stapling instrument of claim 1, wherein the at least one locking notch is configured to receive the at least one trunnion when the stapling assembly is not supported by the second jaw.
10. The surgical stapling instrument of claim 1, wherein the second jaw includes at least one trunnion slot, wherein the at least one trunnion slot includes the at least one locking notch and at least one arcuate slot segment configured to receive the at least one trunnion.
11. The surgical stapling instrument of claim 10, wherein the at least one first abutment surface is configured to disengage the at least one second abutment surface when the at least one trunnion is received within the at least one arcuate slot segment to thereby permit closure of the first jaw toward the second jaw.
12. The surgical stapling instrument of claim 10, wherein the at least one trunnion is configured to move into the at least one arcuate slot segment in response to the stapling assembly being supported by the second jaw.
13. The surgical stapling instrument of claim 1, wherein the at least one first abutment surface tapers downwardly in a proximal direction.
14. The surgical stapling instrument of claim 1, wherein the at least one second abutment surface tapers downwardly in a proximal direction.
15. The surgical stapling instrument of claim 1, wherein the at least one trunnion is resiliently biased toward the at least one locking notch.
16. A surgical stapling instrument comprising: (a) a shaft extending along a longitudinal axis to a distal end; and (b) an end effector operatively coupled with the shaft, wherein the end effector includes: (i) a first jaw including: (A) an anvil, (B) at least one trunnion wall, and (C) at least one trunnion protruding from the at least one trunnion wall, and (ii) a second jaw configured to cooperate with the first jaw to clamp tissue, wherein the second jaw is configured to support a stapling assembly, wherein the second jaw includes: (A) at least one locking notch configured to receive the at least one trunnion, and (B) at least one rocker surface configured to engage the at least one trunnion wall when the at least one trunnion is received within the at least one locking notch to thereby prevent closure of the first jaw toward the second jaw.
17. The surgical stapling instrument of claim 16, wherein the second jaw includes at least one trunnion slot, wherein the at least one trunnion slot includes the at least one locking notch and at least one arcuate slot segment configured to receive the at least one trunnion, wherein the at least one trunnion wall is configured to disengage the at least one rocker surface when the at least one trunnion is received within the at least one arcuate slot segment to thereby permit closure of the first jaw toward the second jaw.
18. The surgical stapling instrument of claim 17, wherein the at least one locking notch is configured to receive the at least one trunnion when the stapling assembly is not supported by the second jaw, wherein the at least one trunnion is configured to move into the at least one arcuate slot

segment in response to the stapling assembly being supported by the second jaw.

19. The surgical stapling instrument of claim 16, wherein the at least one rocker surface tapers downwardly in a proximal direction.

20. A surgical stapling instrument comprising: (a) a shaft extending along a longitudinal axis to a distal end; and (b) an end effector operatively coupled with the shaft, wherein the end effector includes: (i) a first jaw including: (A) an anvil, (B) at least one trunnion wall, and (C) at least one trunnion protruding from the at least one trunnion wall, and (ii) a second jaw configured to cooperate with the first jaw to clamp tissue, wherein the second jaw is configured to support a stapling assembly, wherein the second jaw includes: (A) at least one locking notch configured to receive the at least one trunnion, and (B) a pivot block having at least one surface configured to engage the at least one trunnion wall when the at least one trunnion is received within the at least one locking notch to thereby prevent closure of the first jaw toward the second jaw.